

Tomahawk

17315

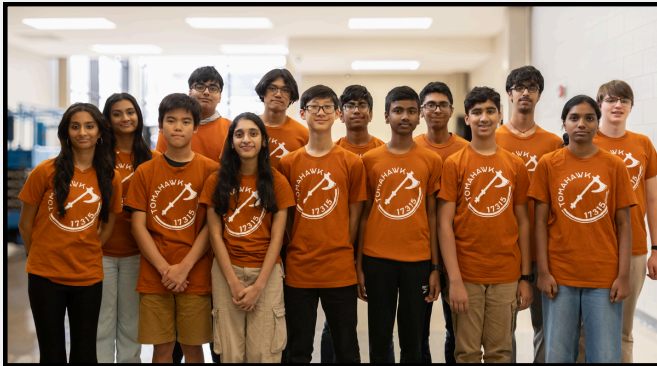


Business Plan

2025-2026



Team Structure



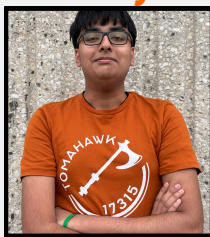
We are team #17315, **Westwood Tomahawk**, from Westwood High School based in Austin, Texas. The 15 of us came together with different backgrounds from rookies to years of experience. In our **5th** year as a competitive team, we seek to empower young minds both locally and internationally by introducing them to FIRST, aid and nurture our fellow FIRST members, and provide others the same opportunities and resources that we are lucky to have access to.

Deekshita



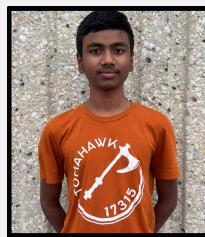
TEAM CAPTAIN

Ritvij



Marketing Lead

Karthik



Hardware Lead

Aditya



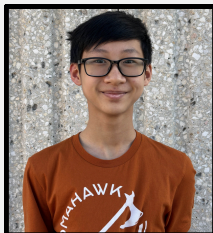
Software Lead

Prahastha



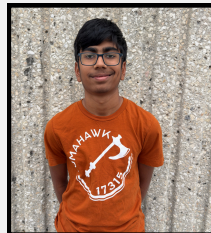
Outreach Lead

Isaac



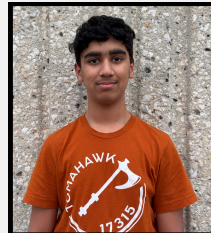
Marketing

Shrikar



Hardware

Minessh



Software

Jusuf



Marketing

Daniel



Outreach

Arjun



Hardware

Saloni



Outreach

Sanaika



Software

Jay



Software

Aron



Hardware



Team Overview

TEAM SUMMARY

FTC Team 17315 originates from Westwood High School in Austin, Texas. In 2020, Team 17315 became the 3rd addition to the Westwood Robotics Organization, which now encompasses a total of four teams. Team 17315 has been participating in FIRST for a total of five years

This year, the team is composed of **fourteen high school students** (three juniors, seven sophomores, and four freshmen), with a rookie to veteran ratio of 4:10.

Each member specializes in one of four subcategories (**three marketing, four software, three outreach, and three hardware**), but all members are highly encouraged to contribute to every aspect of the team.

This helps foster a **healthy learning environment** and enables us to approach problems from multi-faceted viewpoints. Team members are committed to inspiring excellence in our peers and spreading the core values of FIRST within the community and around the world.

Team Motto

"To Forge Our Path"

"To Forge Our Path" embodies Tomahawk's dedication to using purpose, perseverance, and unwavering effort to shape our own success. Every member of Team 17315 is molded by the effort we put into our designs, outreach, and community, just as a tomahawk must be forged with heat, pressure, and precision. **Tomahawk never waits for opportunities; instead, we create them, as evidenced by the concept of "forging our path".** We take on seemingly insurmountable challenges, surpass expectations, and emerge stronger, more capable, and sharper. We work with purpose and impact, whether we're developing FIRST in our community, mentoring less experienced teams, or engineering in the workshop. It represents our conviction that **excellence is developed, honed, and earned rather than accidental.** And in the process, we leave a legacy that encourages others to

Team Overview

Our Organization

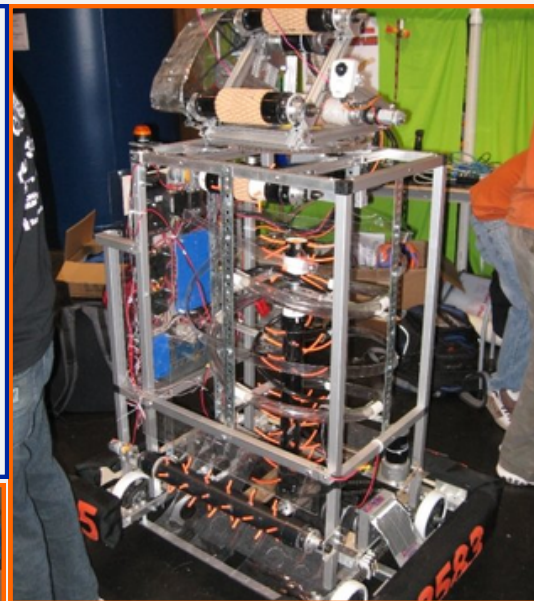
The primary focus of Westwood High School's Robotics Organization is to develop and educate the rising generation of entrepreneurs and leaders in science, technology, engineering, and mathematics (STEM). Here, we promote leadership, teamwork, and mentoring others. Additionally, we aim to encourage students of all ages to pursue STEM initiatives and adhere to FIRST values in order to promote future success.

Westwood Robotics was originally founded in 2000 in Austin, Texas for students wanting to compete in an event called **BEST Robotics**. In **2008**, Westwood Robotics would shift towards FIRST creating **FRC Team 2583 Orange Dynamite** led by **Keith Walker** and **Mr. Mickel**, the club sponsors at that time. Finally, in 2019, **Bailey Hulsey**, the club sponsor at that time helped push for the **creation of 6 FTC Teams**; Teams 17113, 17263, 17315, 18884, 18886, and 20240. In the present day, we maintain 4 FTC teams and 1 FRC Team.

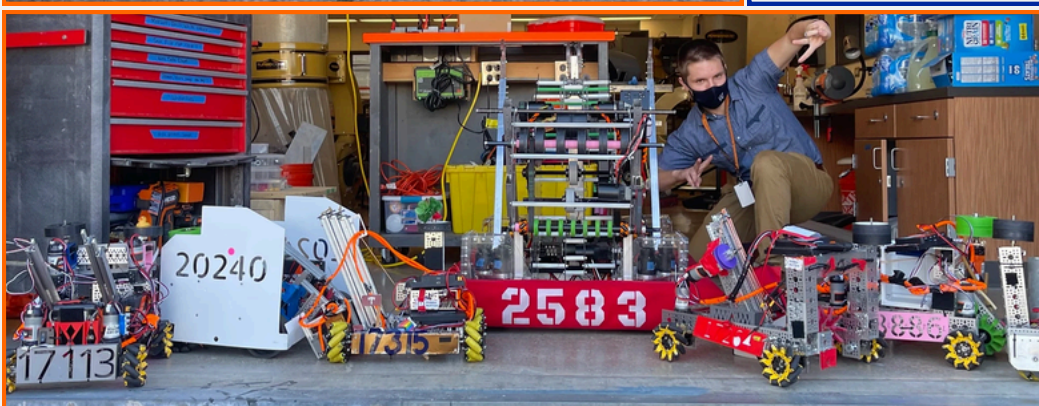
Although the teams work independently on building and programming the robot and constructing the associated marketing materials, they do collaborate and demonstrate "gracious professionalism" by sharing resources, ideas, and coming up with game strategies. This sort of environment mimics the real-world STEM industry, and enriches everyone's experience as a result.



(Left) Our club participating in the BEST Robotics competition back in 2000. This was the start of our club with very few members and little knowledge of robot competitions available to compete in. However, competing in BEST was foundational for the future of the club.



(Top) Our FRC Team Orange Dynamite competing in their first year of existence (2008). This was their first year competing and thus learned a lot from their failures. (Left) our sponsor Mr. Hulsey posing with the robots from the 3rd year of FTC for our club.



Our Team History

2019-2020

This was our first year of existence in which we would go on to advance to the regionals level of competition. **Sadly, this year was cut short for us due to the COVID-19 pandemic.**

2020-2021

The COVID-19 Pandemic deeply hurt our meeting hours and we would thus lose at the district level due to a lack of meeting times and self motivation to work together.

2021-2022

At this point, the majority of members from the last two years had left leading to a major knowledge “vaccum” to take place leading many of our newer members for this year to struggle to compete due to an overall lack of support and knowledge. This year however, we were able to learn many robot related skills which were essential for our knowledge for the future.

2022-2023

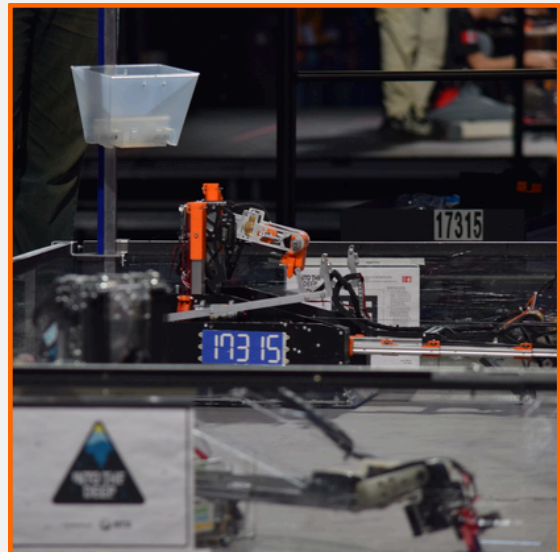
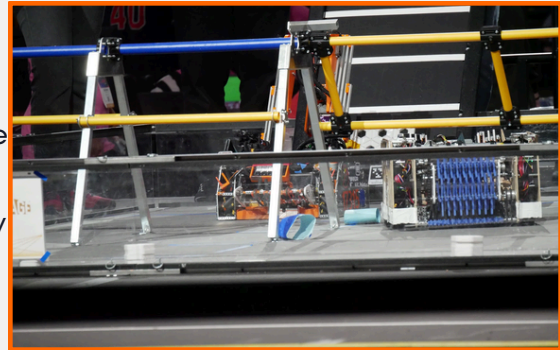
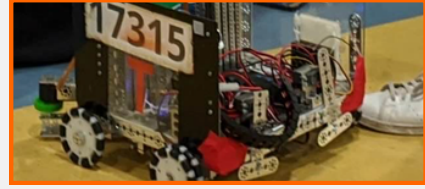
This year we were able to rebound in a knowledge standpoint with many of our members being able to learn more from our successes and failures from the previous year. However, since the competition expanded a great amount, we were unable to advance once again. However, we were able to understand our mistakes and were ready to correct them the next year.

2023-2024

The Centerstage FTC Season was by far the best season in our team’s history. This year, we learned from our mistakes in the previous years and trained many of our teammates over the offseason to ensure they would be prepared for the upcoming season. Although we started the season rough going **1-4 league meet 1**, we were able to rebound, leaning for consistency rather than a higher scoring robot which eventually led us to make the World Championship where we would eventually **win the Edison Division.**

2024-2025

The Into the Deep season was one of our most challenging but most incredible seasons yet. **Last year, 12 of the 13 members from the team the year prior decided to retire from FIRST or join a different team.** The team only had 4 veterans in total when the season began. In the early season, the team was once again constantly losing and being unable to win games. However, as the season progressed, the team improved more and more till eventually at the **Central Texas Semi Area Championship**, Tomahawk was able to be the **Winning Alliance** and then after a 1 month delay, was able to be the **Winning Alliance Captain** at the **Central Texas Area Championship** due to a rebuild during delay. Although Tomahawk didn’t win their division, but given their early struggles and lack of experience, the season was considered a success.



SWOT Analysis

WHAT IS SWOT?

SWOT stands for **Strengths, Weaknesses, Opportunities, and Threats**. It is crucial for teams to create their own SWOT Analysis so that they identify potential issues that should be avoided, as well as see their potential opportunities to improve and overcome weaknesses.

STRENGTHS

Ambition: Due to our success in past seasons, our team is now more motivated than ever to succeed in this year's Decode challenge. Whether it be excelling in competition or having the opportunity to spread our passion for STEM within our community, we have set goals for our team that we are excited to work to meet.

Experience: Last season, our rookies learned valuable life-skills and worked all year to strengthen them. At the same time, our veterans developed more self-confidence, along with skill. With many of our members on their second, third, and even fourth year of FTC, our veterans are eager to employ their newfound skills and productive energy into this season.

WEAKNESSES

Finances: We lack the funds sufficient to fund major outreach projects. In order to start and sustain such projects, it is essential we begin to fundraise as soon as we can.

Commitments: Last year although we were able to do well, we still saw a lack of commitment by the majority of the team, and issue we hope to counter this year by creating a calendar with many of the dates of our events as well as outlining commitments at the start of the yr of brief expectations we hold everyone to who participate on the team.

Poor Documentation: In past years we have struggled to document our robot journey well. We hope that this year we are able to document better due to the mentorship and guidance we recieved from meeting with industry professionals.

OPPORTUNITIES

Resources: Although we lack resources for documentation, we are still given a space to work by our school as well as parts from last year and the support of a school club which includes machinery to aid us to create a well functioning robot. We are also given references from previous years teams in our organization to help us better learn and improve upon their work.

Community: From our learnings from last year, we realized that there are ample opportunities both available in our community and worldwide to be able to spread FIRST and robotics to. We also learned that there were many companies that we could reach out to. This would help us learn and incorporate industry design into our own robot construction.

THREATS

Time Restriction: The UIL time restrictions hinder our ability to work on the robot for more than 8 hours each week. This will not be a problem, however, because we will make sure to make the most of our time in the lab.

Grades: In the Westwood Robotics program, there is an expectation to keep our academic studies as a priority, as it can be easy to dismiss schoolwork when focused on our robots. As a rule, if a student's class average in any class falls below 80%, that student is placed on probation. During this time, many Westwood Robotics students will often step up to tutor this student and help them raise their grade. Once elevated, the student is taken off of probation. However, if a student is on probation and does not improve their grade, they risk dismissal from the program.

Team Organization



TEAM STRUCTURE



Leadership Council

SUBTEAMS

Adult
Mentors

Team Captain
Deekshita

Marketing
Subteam
Lead: Ritvij

Software
Subteam
Lead: Aditya

Hardware
Subteam
Lead: Karthik

Outreach
Subteam
Lead: Prahastha



Tomahawk
Members

Marketing
Team
Members

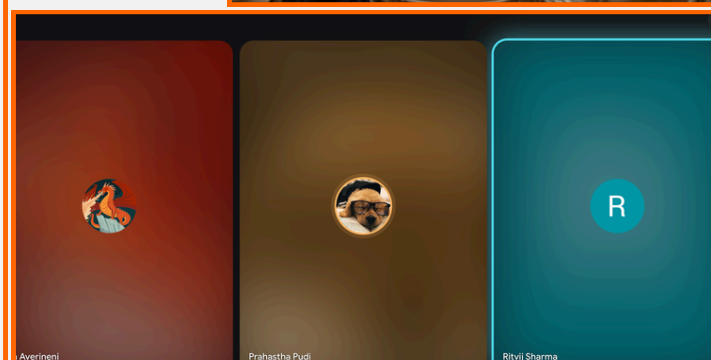
Software
Team
Members

Hardware
Team
Members

Outreach
Team
Members

Taking Inspiration from Boy Scouts

Our type of leadership and communication was heavily inspired from the Boy Scouts PLC System. In the Boy Scouts PLC system, student troop leaders will speak together monthly with adult leaders in attendance. The most important part of the PLC system is that adult leaders do NOT lead the discussions, rather let the student leaders discuss their issues and only intervene when they deem it is needed. Similarly, in our meetings adult mentors are simply there to observe and give suggestions where they feel it is necessary. This allows the team to be student driven and makes our passion for FTC greater. Also in the PLC Meeting, Troop(Boy Scout) Members are allowed to request an issue to be brought up at the meetings to anyone on the leadership crew. If they feel they themselves need to bring up the issue, the Scout is welcome to attend. Similarly, we have a system where a team member can request a topic to be brought up or may attend the leadership meetings we hold monthly and discuss it with us.



RECRUITMENT

Each school year in April, after the competition season has ended, the Westwood Robotics Organization posts flyers around the school at the very start of the school year to advertise robotics. Many times, teams will also **recruit within the school** asking other robotics members if they wish to join their team for the upcoming year. This recruitment **mix of both veterans and new rookies, allow for the team to stay sustainable** and continue to be able to add new talent when veterans retire or graduate as well as add new experienced members from other teams who may be able to contribute and guide newer members. Students apply by **filling out an application that will be reviewed by the leadership of each team** at the start of the year. Appropriate members are identified using the following guidelines:

Commitment	While it is not necessary for prospective members to know everything coming in, it is important that every member has the time to dedicate to the robotics season. We encourage hands-on experience that can only be accomplished if everyone is sufficiently committed . With that being said, we want every member on our team to have the option to explore other options, and we work as a team to manage conflicts and work around everyone's schedule. To work around this, we work within UIL constraints to coordinate meetings within subgroups.	1
Process	Our organization has a selection process designed to ensure that everyone in our program is committed to robotics and will benefit from being there. After members are admitted, project managers are in charge of putting together a team, with a focus on diversity and varying range of experience and interests. The goal is to give every member of the team a clear role they enjoy, but also the chance to grow and learn from others.	2
Academic Success	While robotics is important and a passion for many of us, it is important to place academics first . To participate in robotics, members are required to maintain a 3.0 (out of 4.0) unweighted average and keep all grades above 80 to participate. Our robotics program also has a system in place to help people who are struggling in a class. We have a lot of smart people across all grades who take the most challenging classes at our school, and, as a result, peer tutoring is made available for anyone who needs it.	3

BUDGET

This year in order to be financially responsible with our money, we decided to create a sheet with not only our expenses, but **our income to plan and pay for it**. We hoped by keeping track of our finances, it will make us think more carefully about the value that certain purchases would bring and allow us to truly see if the **impact vs cost** is truly **great enough to justify buying it**. This was also critical as in our efforts to fundraise, we learned about the **difficulty to fundraise** which led us to creating this **list of expenses and incomes to help us stay better organized** in our efforts to have enough money for the entire season.



EXPENSES



ITEM	PRICE	EXPLANATION
VEX Robotics Parts	\$45	Needed vex flywheel weights to make our rotational inertia greater
Filament	FREE	Sponsored by Polymaker
Wiring and Electronics	\$50	Wiring Materials
Wood	\$60	The material we used to prototype our robot parts
Aluminum	\$20	Used to create more permanent, less changable parts such as turret plates
Motors and Servos	FREE	Given as part of Westwood Robotics Membership
Marketing Supplies	\$285	Posters and Pit Decorations
Practice Field	FREE	Given as part of Westwood Robotics Membership
Glowforge, CNC, and other machines access and maintenance	FREE	Given as part of Westwood Robotics Membership
Dubai Outreach Program Expenses	\$1000	Price to send aid to UAE, raise awareness of the initiative
Most Channels, Screws, and general parts	FREE	Given as part of Westwood Robotics Membership
Controllers	\$100	Controllers with less drift and optimized connection to driver station

TOTAL PRICE: \$1560 to date

BUDGET

This year in order to be financially responsible with our money, we decided to create a sheet with not only our expenses, but **our income to plan and pay for it**. We hoped by keeping track of our finances, it will make us think more carefully about the value that certain purchases would bring and allow us to truly see if the **impact vs cost** is truly **great enough to justify buying it**. This was also critical as in our efforts to fundraise, we learned about the **difficulty to fundraise** which led us to creating this **list of expenses and incomes to help us stay better organized** in our efforts to have enough money for the entire season.



EXPENSES



ITEM	PRICE	EXPLANATION
FTC Registration Fee	\$295	Season Registration Fee
First in Texas Registration Fee	\$275	Season Registration Fee
Westwood Robotics Membership Fee	\$1500	Fee for our team to be part of the Westwood Robotics program
T-shirts	\$200	Team Shirts for Team Members
VEX Robotics Parts	\$45	Needed vex flywheel weights to make our rotational inertia greater
Filament	FREE	Sponsored by Polymaker
Wiring and Electronics	\$50	Wiring Materials
goBILDA Parts	\$800	Specific GoBilda parts not found in Makerspace
Wood	\$60	The material we used to prototype our robot parts
Aluminum	\$20	Used to create more permanent, less changable parts such as turret plates
Motors and Servos	FREE	Given as part of Westwood Robotics Membership
Marketing Supplies	\$285	Posters and Pit Decorations
Practice Field	FREE	Given as part of Westwood Robotics Membership
Glowforge, CNC, and other machines access and maintenance	FREE	Given as part of Westwood Robotics Membership
Dubai Outreach Program Expenses	\$1000	Price to send aid to UAE, raise awareness of the initiative
Philippines Initiative	\$700	Expenses for paying for sending parts and support to the Philippines
Most Channels, Screws, and general parts	FREE	Given as part of Westwood Robotics Membership
Controllers	\$100	Controllers with less drift and optimized connection to driver station

TOTAL PRICE: \$5145 to date

BUDGET



INCOMES



ITEM	Money	EXPLANATION
Gene Haas Grant	\$2000	Gene Haas Grant Received
Polymaker Grant	N/A	Free Filament for the entire season
JukeBox Sticker Grant	N/a	200 dollars worth of JukeBox stickers products
Cupprimo Sponsorship	\$150	150 dollars for our robot and free cupcakes for competitions
FLL Tutoring	\$50/per 2 hour session	Tutored kids on how to make working FLL robots
Swyft Sponsorship	\$200	voucher for anything from Swyft
FIRST in Texas Grant	\$500	500 dollars from First in Texas
Qualcomm Grant	\$500	500 dollar grant awarded from Qualcomm

TOTAL PRICE: \$3650 to date

This year we not only hoped to stay **sustainable** in our membership but also our **finances**. By not only applying for major grants but by presenting to local sponsors(more on the next page). However, we also decided to **start our FLL student tutoring program** as we realized we wanted a **constant source of income** due to the disadvantages that grants and even pitching to local companies **wasn't sustainable year around**. Additionally, many grants had a deadline and pitches take a **great deal of time to set up**, time which would be better spent for **presentation and portfolio** later on in the season.

BUDGET



INCOMES



ITEM	Money	EXPLANATION
Team Dues	\$3000	\$200 dues per team member
Gene Haas Grant	\$2000	Gene Haas Grant Received
Polymaker Grant	N/A	Free Filament for the entire season
JukeBox Sticker Grant	N/a	200 dollars worth of JukeBox stickers products
Cupprimo Sponsorship	\$150	150 dollars for our robot and free cupcakes for competitions
FLL Tutoring	\$50/per 2 hour session	Tutored kids on how to make working FLL robots
Swyft Sponsorship	\$200	voucher for anything from Swyft
FIRST in Texas Grant	\$500	500 dollars from First in Texas
Qualcomm Grant	\$500	250 dollar grant awarded from Qualcomm

TOTAL PRICE: \$6670 to date

\$5145

Total Funds
Required

\$6670

Raised
(excluding
future income)

\$1275

total left over

This year we not only hoped to stay **sustainable** in our membership but also our **finances**. By not only applying for major grants but by presenting to local sponsors(more on the next page). However, we also decided to **start our FLL student tutoring program** as we realized we wanted a **constant source of income** due to the disadvantages that grants and even pitching to local companies **wasn't sustainable year around**. Additionally, many grants had a deadline and pitches take a **great deal of time to set up**, time which would be better spent for **presentation and portfolio** later on in the season.

SPONSORSHIP STRATEGY

Emails

- Our original approach to fundraising
- Targeted toward local businesses as well as major cooperations
- Realized from experience to target local businesses more for a better reply chance
- Strategy: send out as many as possible
- Low response rate ~1% for us this year
- Less chances of higher financial commitment from companies

Grants

- Targeted toward **tech companies and foundations**
- Fill out an Application and wait for a response
- Takes more time than an email **BUT** usually offers higher upside on total donated or incentives given
- Makes up the **majority** of our **funds**
- However, isn't **sustainable** as grants close by a certain deadline

Pitches

- Learned method from **18139 Rebel Robotics**
- Emails are to set up pitches
- Allows us to **be more personal** with the companies
- So far have had a **90%** success rate with this method in landing sponsorships
- MUCH **higher success rate** compared to simple cold emailing
- Is the most **sustainable** fundraising method

Our Partners



FLL Tutoring

One way we wanted this year to be **financially sustainable** was through a constant income. Due to many companies and grants only being open for a certain time and only giving a set amount, it was much riskier to acquire and would eventually become difficult later in the season to do so due to **restrictions from the companies** we reached out to and **deadlines from grants**. These constraints along with the fact that **sponsorship pitches had a relatively low success rate** led us to rethink our finances. Instead, we came up with an idea **inspired from robocamp**: Offer classes for students on how to program FLL level robots and help them improve both their **building skills** and their **programming skills(Python)** to help spark their interest in FIRST and **eventually to join the program** with an advantage using the **skills they had learned**.

Spike LEGO Tutoring

Members of our team tutor Barakh and Veda in person weekly. By teaching basic programming language like block programming and building with the Spike LEGO kit, we also give them an introduction to FLL. Through the process of building, finding mistakes, innovating and redesigning the build, the kids problem solving skills improved. and that made this process meaningful. Additionally, our lesson plan includes CAD and scratch.



Quotes from some of the families

"Over the span over 5 weekends, Barakh has learned fundamental programming concepts such as conditionals, motor controls, and loops in the SPIKE coding interface." – Barakh's family

"By working with the LEGO Spike kit, Veda has developed a natural understanding of the essential components of a robot and the overall design process." – Veda's family

By earning money from weekly tutoring, we were able to make a lot of money. **All of the money goes to buying supplies, including parts for the robot, as well as to supply starter kits and other materials to FTC teams we started in the Philippines and UAE.**

We found out that by having a **constant income**, we were able to **smoothly manage our budget**, instead of relying on grants and pitches.